

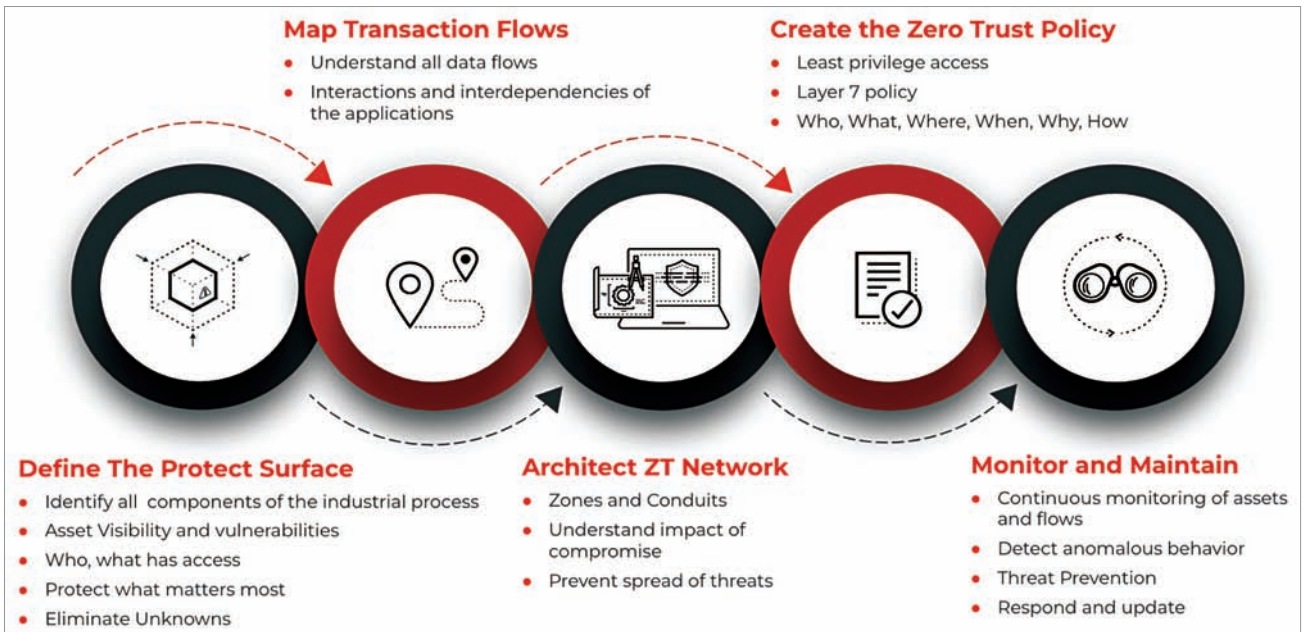


Fig. 5: Zero trust security model

or expediency can lead to vulnerabilities.

As organisations forge ahead with digital transformation, they must

prioritise security in tandem with innovation. A robust security strategy, modernisation of legacy systems, and ongoing monitoring and improve-

ment efforts are critical for protecting Industrial IoT assets. By addressing these challenges proactively, organisations can navigate the perfect storm of threats in the digital transformation era and emerge stronger.

In this article, we delve into the strategies and principles that can fortify your IIoT security posture.

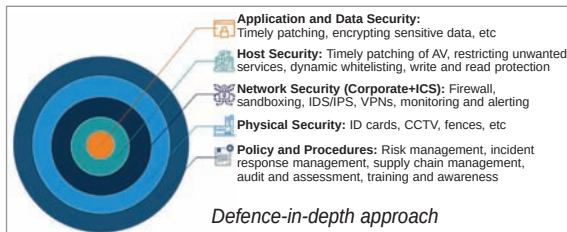
## Pillars and automation in IIoT security

To build a robust IIoT security framework, focus is required on three key pillars: people, process, and technology. Recognising that humans are often the weakest link in security, investment should be done in security awareness training, gamify it for employee engagement, and conduct thorough background checks for new hires. Security by design is vital. It is necessary to establish clear processes, governance, and a comprehensive security strategy.

One needs to define the security architecture, including essential security controls such as firewalls, endpoint security, SIEM (security information and event management),

## A defence-in-depth approach

Implementing a defence-in-depth approach is crucial to safeguard IIoT environments effectively. This strategy mirrors the layers of defence employed in the US military, where multiple layers of protection deter and thwart adversaries at various levels. In the context of IIoT security, consider these layers:



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- **Application and data security.** Prioritise timely system patching and encrypt sensitive data.
- **Host security.** Deploy endpoint security tools, security development lifecycle (SDL), and application development security response (ADSR) mechanisms.
- **Network security.** Utilise traditional security measures such as sandboxing, VPNs, and firewalls.
- **Physical security.** Implement access controls, ID card systems, and surveillance to protect physical assets.
- **Policies and processes.** Develop and enforce robust security policies, procedures, and incident response plans.
- **Compliance with standards.** Consider adhering to the IEC 62443 standard, a gold standard for OT security, covering terminologies, concepts, glossaries, policies, procedures, system protection, and security risk assessment.

By integrating these layers of security, IIoT environments can effectively minimise risks, protect critical infrastructure, and ensure the resilience of industrial operations. It is imperative for organisations to recognise the urgency of adopting such a holistic security strategy in safeguarding the future of IIoT.